SAIHAJVIR SINGH

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SUMMARY

Motivated B.Tech Computer Engineering student (2027) with strong foundations in software development, full-stack development, and machine learning. Proficient in Java, Python, C++, JavaScript, React.js, Node.js, MongoDB, and Firebase, with hands-on experience building scalable web, mobile, and AI-based applications. Developed projects including a CNN-based wildfire detection model achieving 95% accuracy and real-time Android applications with cloud integration. Strong knowledge of data structures, object-oriented programming, and problem-solving, with proven leadership and teamwork experience. Seeking opportunities to apply technical skills in real-world software and technology projects.

PROJECTS

Wildfire Detection System (Deep Learning) — July 2025 - Dec 2025

Technologies: Python, CNN, Deep Learning, OpenCV, TensorFlow

- Developed a CNN-based wildfire detection model using 4-channel input (RGB + Thermal imaging) to identify wildfire-prone regions.
- Implemented image preprocessing techniques, including normalization and augmentation, to improve model performance.
- Designed and executed the training and validation pipeline, achieving 95% accuracy on the dataset.
- Optimized model performance for early wildfire detection to support environmental monitoring and disaster prevention.

Smart Parking Finder (Android Application) — July 2025 - Dec 2025

Technologies: Java, Firebase, Android Studio

- Developed an Android application to display real-time parking slot availability using Firebase cloud database.
- Integrated user authentication and real-time data synchronization for accurate parking updates.
- Designed an intuitive user interface to enhance accessibility and improve user experience.
- Implemented live data updates to minimize parking search time and improve system efficiency.

AI Blackjack Game — July 2025 - Dec 2025

Technologies: Python/Java (use whichever you used), Probability Logic

- Designed an AI-based Blackjack game implementing probability-based decision strategies.
- Developed logic to simulate intelligent gameplay behavior based on game conditions.
- Implemented game rules and scoring system to ensure realistic gameplay simulation.
- Enhanced understanding of AI decision-making and probability algorithms.

Gym Management System — Jan 2025 - July 2025

Technologies: Java, Database (SQLite)

- Developed a Java-based management system to handle memberships, attendance tracking, and payment records.
- Implemented CRUD operations to manage user data efficiently.
- Integrated database connectivity for secure and structured data storage.
- Improved record management efficiency through automation of manual processes.

VANRA – Blue Carbon Registry (Smart India Hackathon) — July 2025 - Dec 2025

Technologies: TypeScript, Next.js, Supabase, JSON, HTML, CSS

- Contributed to the development of VANRA, a digital platform for tracking and registering coastal carbon credits during the Smart India Hackathon (SIH).
- Built responsive frontend components using Next.js and TypeScript to manage environmental data and user interactions.
- Integrated Supabase backend for authentication, database management, and real-time data storage using structured JSON records.
- Collaborated in a team environment to design scalable modules for carbon credit tracking and environmental monitoring.

EDUCATION

Bachelor of Technology in Computer Engineering **July 2023 - Aug 2027**

- Mukesh Patel School of Technology Management & Engineering
- CGPA: 3.02 / 4.00

HSC (CBSE) **July 2021 - June 2023**

- Percentage: 77%

SSC (CBSE) **July 2020 - June 2021**

- Percentage: 89%

CERTIFICATIONS

- **Python Basics-** University of Michigan (Coursera)
- **Python Data Structures-** University of Michigan (Coursera)
- **Programming for Everybody-** University of Michigan (Coursera)
- **Introduction to Front-End Development-** Meta (Coursera)
- **Full Stack Development-** Meta (Coursera) (In Progress)
- **Research Paper Published-** SmartCom Conference, Pune

CORE SKILLS

Technical Skills:

- **Full Stack Development:** HTML, CSS, JavaScript, React.js
- **Backend Development:** Node.js, Express.js
- **Database:** MongoDB, Firebase
- **Programming:** C, C++, Java, Python
- **Core Concepts:** Data Structures, OOPJ
- **Tools & Technologies:** Android Studio, Firebase Console, GitHub, VS code

SOFT SKILLS

Strong leadership and team collaboration abilities, strategic thinking, effective problem-solving, analytical mindset, clear communication skills, and efficient planning & execution.

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

Head of Photography Department - IET MPSTME (2024-25)

- Led and managed a team of photographers for college events, ensuring high-quality visual documentation and timely content delivery.
- Coordinated event coverage, task allocation, and creative planning to maintain consistent output standards.
- Strengthened leadership, teamwork, and project coordination skills through effective team management.

Senior Executive - Outreach Department, Student Council (Sattva) (2025-26)

- Managed outreach initiatives and supported communication strategies to enhance student engagement and participation.
- Coordinated with internal teams to organize events and streamline outreach operations.

Subhead - Outreach Department, The Editorial Project (2025-26)

- Assisted in planning outreach activities and maintaining communication with collaborators and contributors.
- Supported team coordination and execution of content-related initiatives.

Executive - Corporate Relations, Analytika (2023-24)

- Assisted in establishing and maintaining partnerships with external organizations and sponsors.
- Supported communication and coordination for professional events and networking initiatives.

Volunteer - Hunjan Hospital, Ludhiana (2024)

- Participated in community service activities supporting hospital operations and patient assistance initiatives.
- Demonstrated teamwork, responsibility, and commitment to social service.

Participant - Smart India Hackathon (SIH) | Project: VANRA – Blue Carbon Registry (2025)

- Collaborated in a team to develop a digital platform for coastal carbon credit tracking and environmental monitoring.
- Contributed to solution design and technical development during the national-level innovation challenge.